Introduction to Remote Sensing



Remote: not in contact with, Away from

Sensing: The act of observation

- It is the acquisition of info about an object, area or phenomenon while still at some distance from it, (not in contact)
- The technique of obtaining info about objects through the analysis of data collected by special instruments that are not in physical contact with the objects of investigation, can be regarded as ***reconnaissance from a distance***
- Its is the art or science of telling something about an object without touching it.

The act of seeing is in effect "*remote sensing*" and eye is a "*remote sensor*"

Remote Sensing

Mapping: collection of thematic and quantitative baseline data

Measuring: more rigorous mapping process by quantifying and documenting attribute of phenomena

Modeling: process of describing a system under study through precise and typically mathematical relations of inputs and outputs, and to simulate its present, past or future behavior

Monitoring: regular assessment of the conditions by recording the changes in natural phenomena and human activities

These functions require valid, up-to-date, spatially and temporally explicit natural resource data for decision-making

RS History

In 1860s, Photographs were taken from balloons

In 1890s, Kites and trained pigeons, were used for photographs

1900 –1910s, photographs were taken from airplanes

1914-1918, Aerial photography became effective during world war-I

1930 –1940s, Germany developed radar tech and IR film was develop during the war-II

From 1957, onwards remote sensing by earth orbiting satellites started

RS of the Earth

1957 – sputnik—soviet union-start of space age, turned the thinking to consideration of how the satellite might be useful

1958 – national space act-NASA was formed, Could aerospace technology be used to better manage the Earth's natural and man made resources?

1960 – tiros-1,basically a weather sat, principally to provide image of clouds cover over wide area, first use of term "remote sensing"

1960 - 1980 – some 40 earth observational satellites launched (Landsat, SPOT)

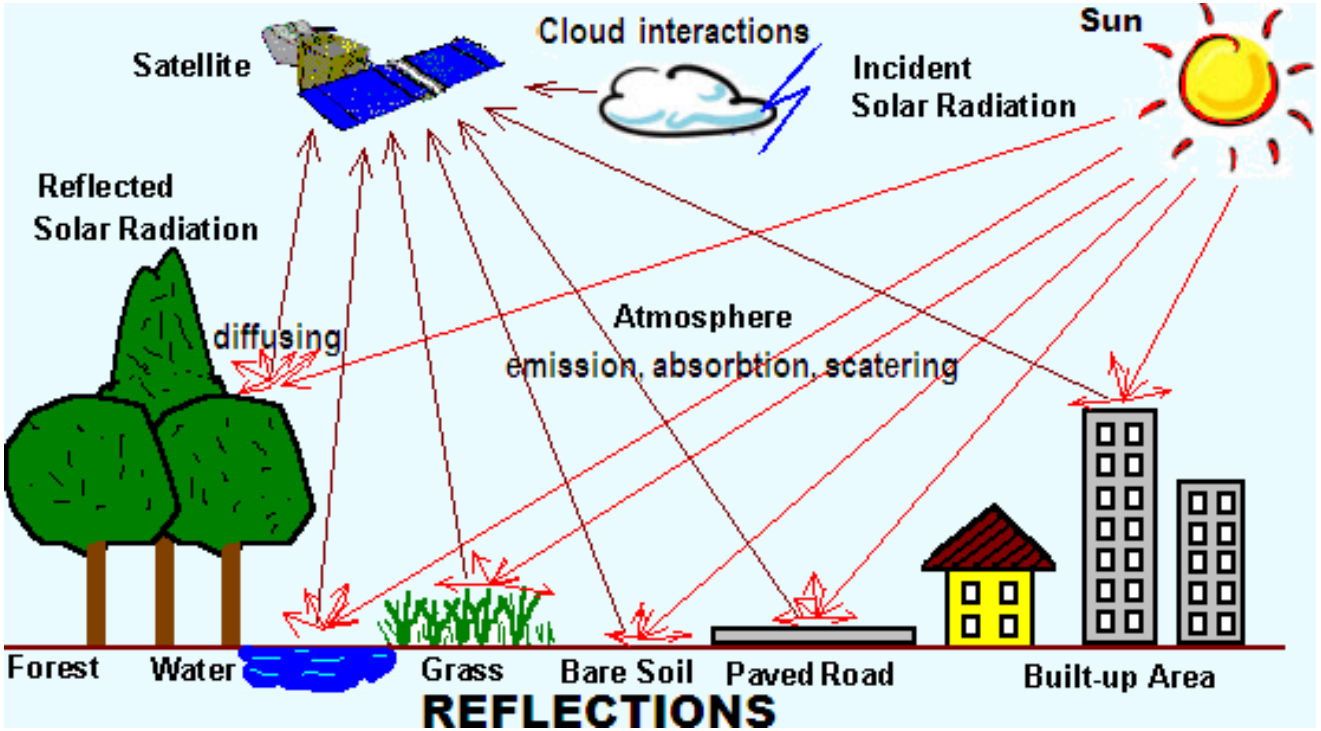
Late 1990s, high resolution commercial satellites (IKONOS)

2000-2010, new era of very high resolution sat- QuickBird, orb view, GeoEye,WorldView

Principle

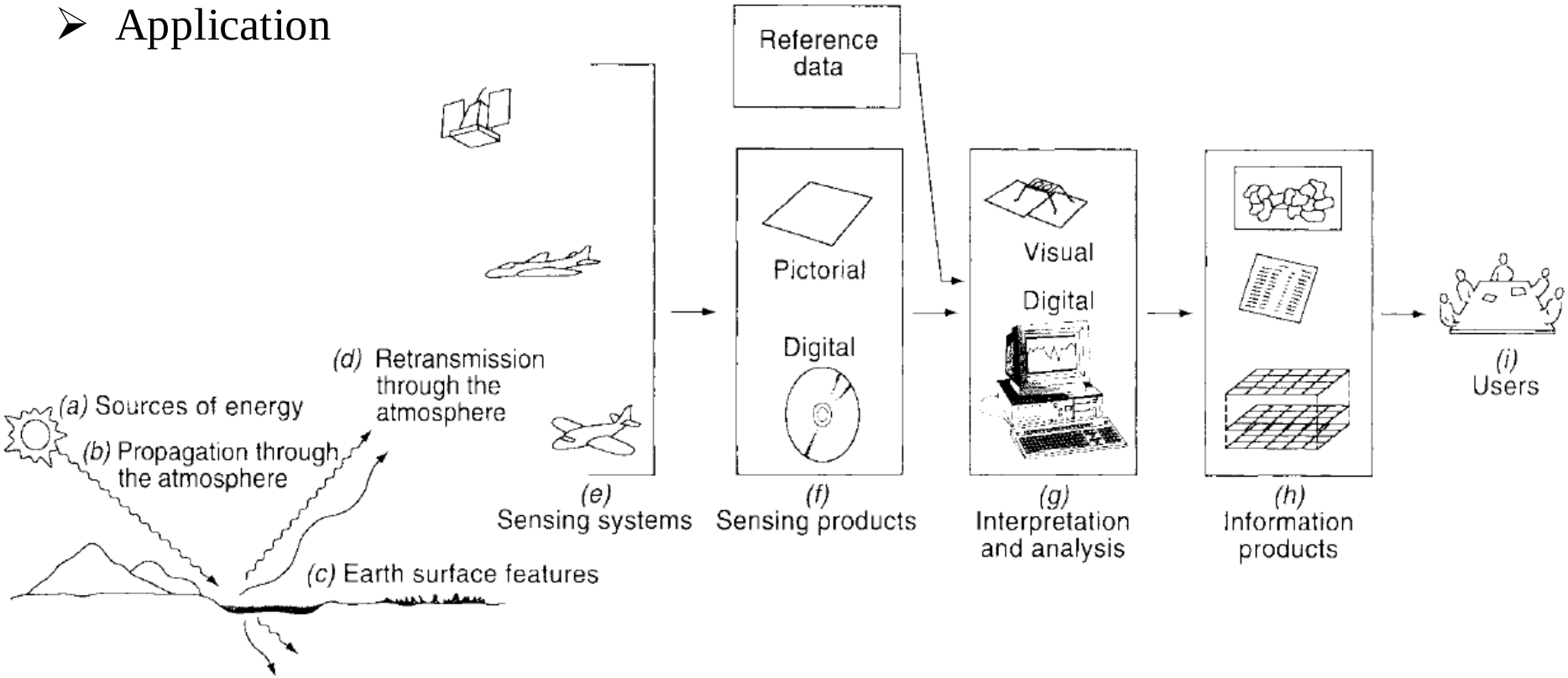
RS largely concerns the measurement of Electromagnetic energy which is reflected, scattered or emitted by different earth surface objects

Different objects return different kinds and amount of energy, detecting these differences enables their identification



Remote Sensing Process

- Energy Source or Illumination
- Radiation and the Atmosphere
- Interaction with the Target
- Recording of Energy by the Sensor
- Transmission, Reception, and Processing
- Interpretation and Analysis
- Application



Remote Sensor

The detection and recording instruments for RS are known as *remote sensors* and include

- Photographic cameras
- Scanners
- Radar systems

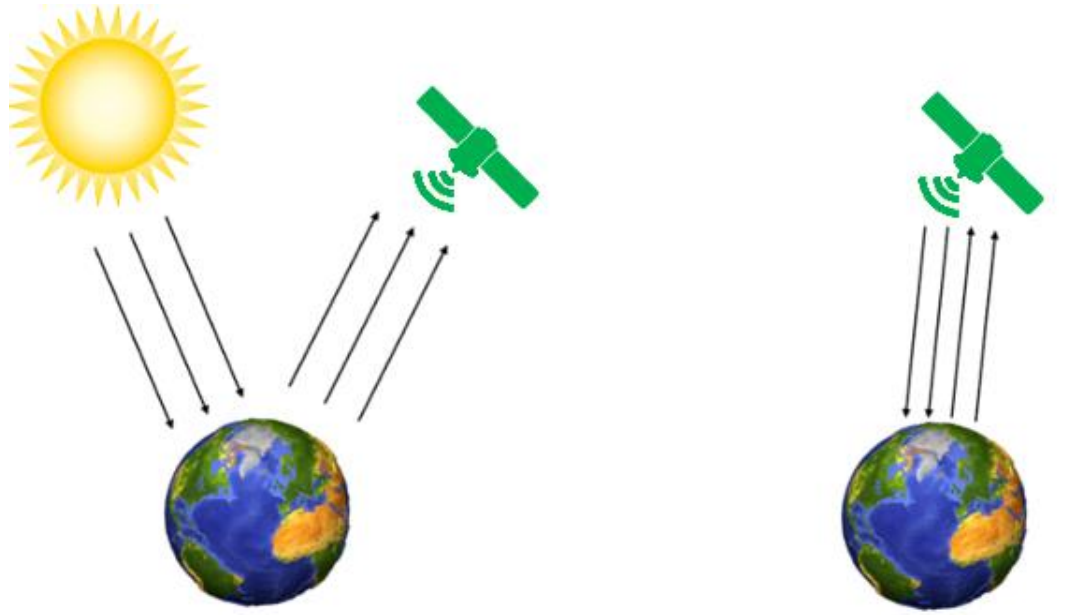
These are carried on aircrafts and earth orbiting space crafts, which has led to the familiar phrase

"*Eye in the sky*"

Passive and Active sensors

Important sensor parameters: Spatial, Spectral and Temporal resolutions

Partially under human control (*Design, Operation*)



Passive RS

Active RS

Scene: part of the system which is out in front of the sensor, Earth Surface and the Atmosphere, Not being under human control

Data: Representation that can be operated upon

Information: Data that has been interpreted by human beings

Advantages of RS

- Large area coverage enabling global, regional surveys of variety of themes
- Repetitive coverage, allowing monitoring of dynamic themes like water, agriculture, etc.
- Data acquisition over inaccessible areas
- Data acquisition over multiple heights allowing different scales and resolutions
- Fast and accurate data processing

Limitations of RS

- Often oversold, it cannot provides all the information needed, rather provides some spatial, spectral, and temporal information that we hope is efficient and economical

- Human beings select the appropriate remote sensing system to collect the data, specify resolutions, calibrate the sensor, select the platform, determine when the data will be collected, and specify how the data are processed, this may introduce errors

- Remote sensing instruments may become uncalibrated, resulting in dubious remote sensor data

- Remote sensing data may be expensive to collect and analyze. Hopefully, the information extracted from the remote sensor data justifies the expense

Application of RS

Successful application of any RS system is “ the person or persons” using this data

RS Data is just raw but when processed by a human interpreter, becomes "the usable info” only, and its applications depends upon how a person :-

- ✓ Understands it
- ✓ Knows how to interpret
- ✓ Know how best to use it

A thorough understanding of the problem is essential to the productive application of any RS data

Also, no single combination of the data acquisition and analysis procedures satisfy the needs of all data users

Application of RS

Some of the applications of RS data can be

- Mapping (Topography, Land use/ land cover, Infrastructure)
- Resource Exploration (Geologic and soil mapping)
- Agriculture (crop monitoring and prediction)
- Forestry
- Water resources
- Urban and regional planning
- Wetland mapping
- Natural Disaster Management
- Environments assessment
- Military surveillance and reconnaissance

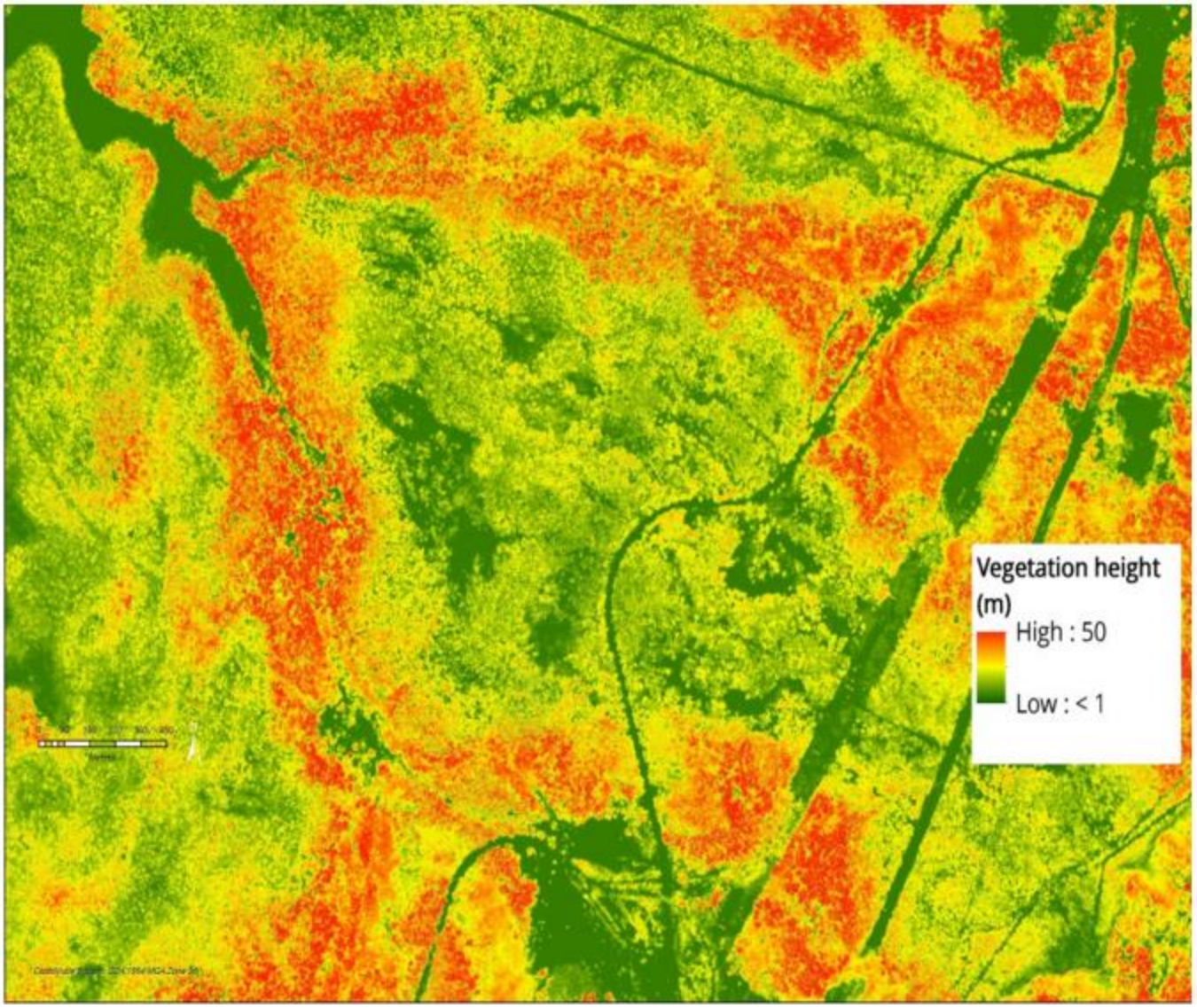
Updating Urban Maps



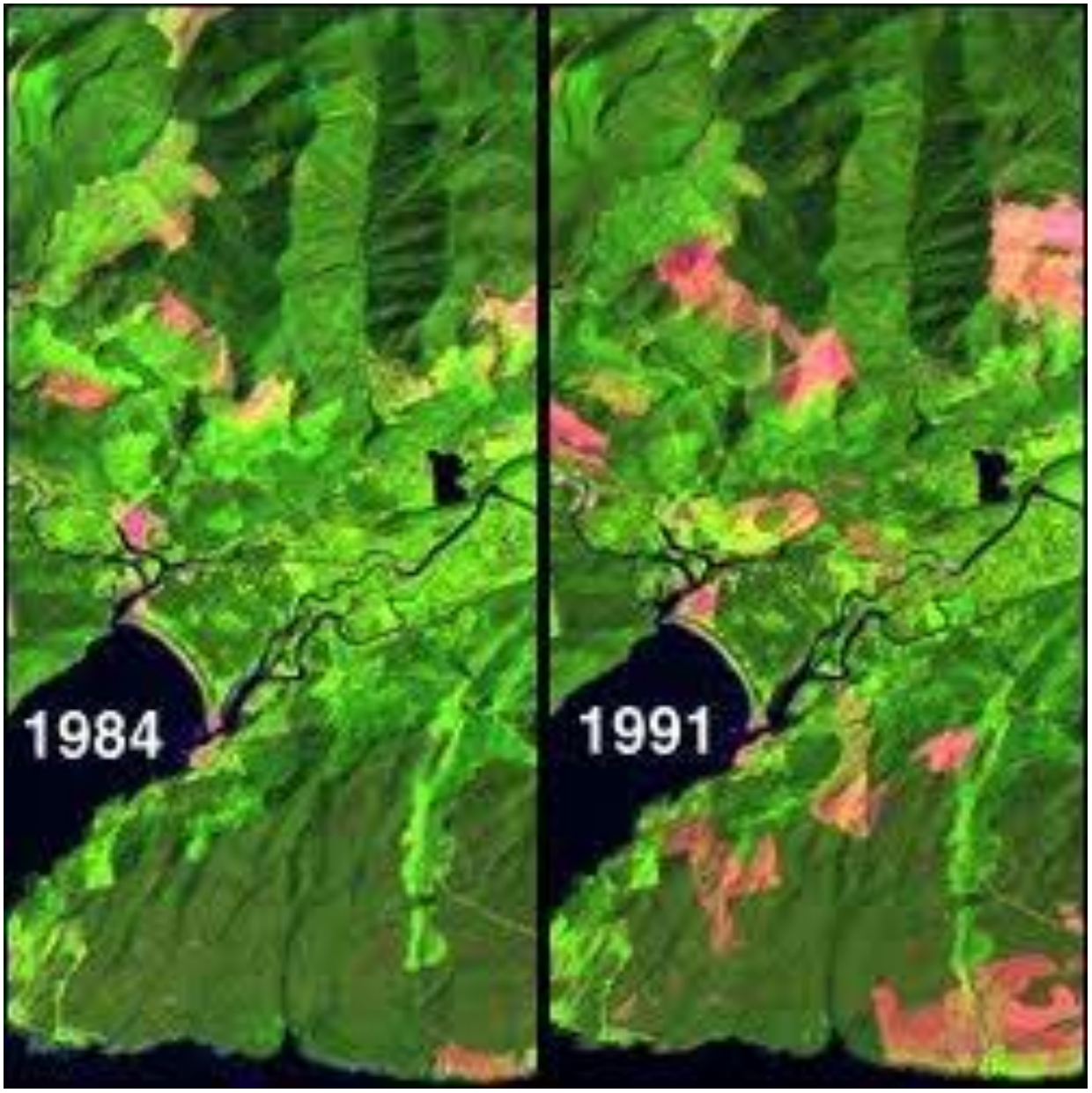
Flood Damage Monitoring



LiDAR analysis showing height of vegetation



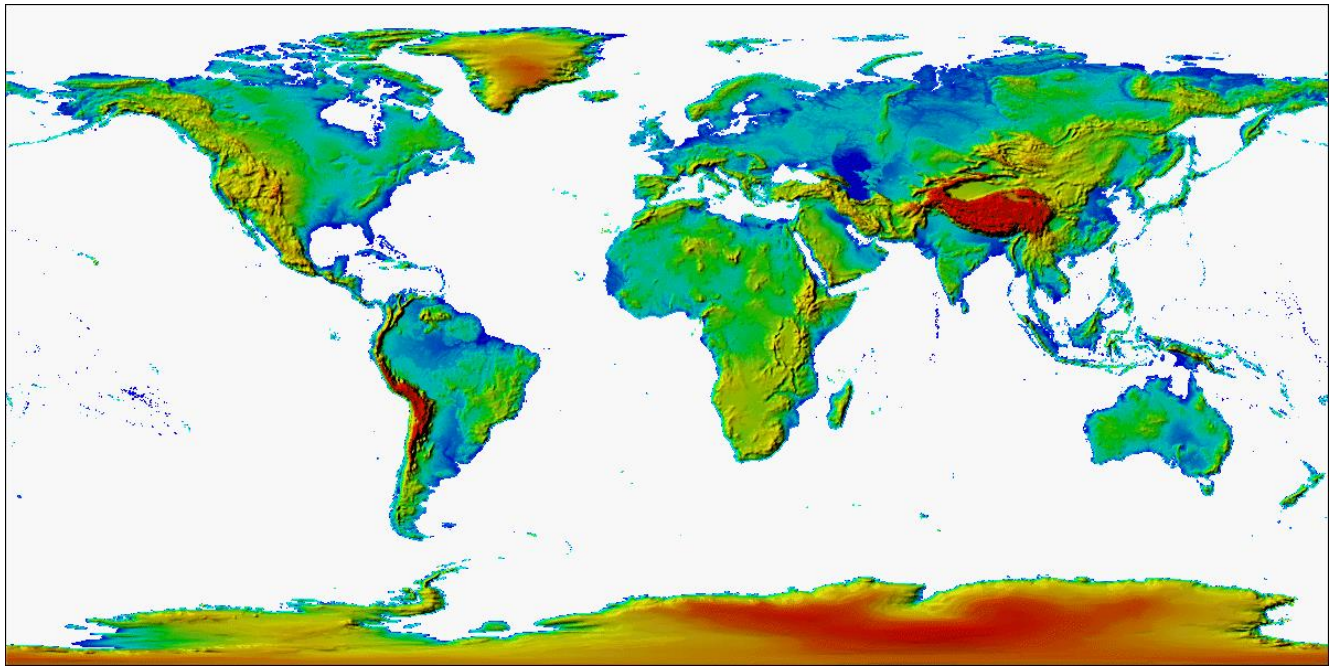
Forestry – Change Detection



Application of RS

It is so broad in its applications that nobody owns the field and benefits are derived by all for basic research or operational applications

RS is and will continue to play an increasingly broad and imp role in inventorying, monitoring and managing earth resources






ANY
QUESTIONS
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